

Applied Forecasting: Stress Prediction Pipeline

13-Model Benchmarking & Analytical Validation

Nilesh Mori | Harsh Maheriya | Dhanush Pillai | Megh Jagtap

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PART I

Introduction & Data Audit

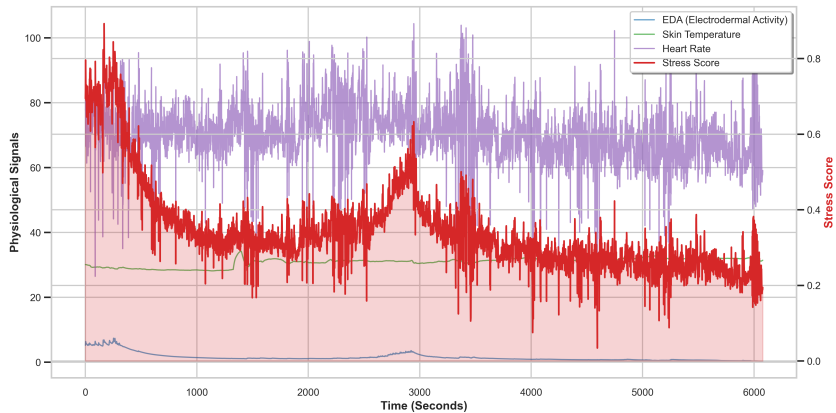
Problem Statement: Proactive Stress Monitoring

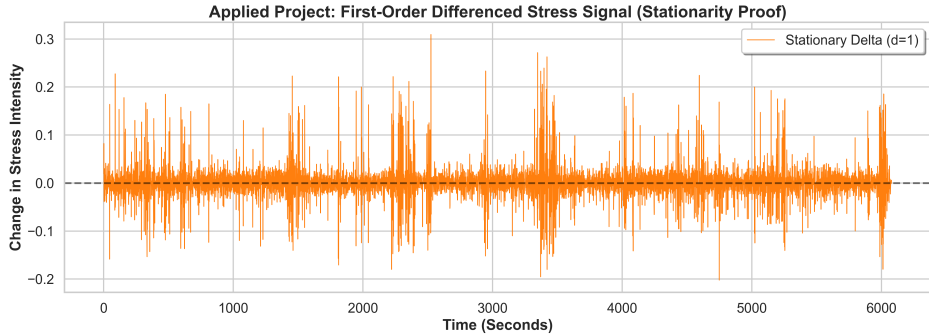


- ▶ **Reactive:** Detecting stress **after** onset.
- ▶ **Proactive:** Forecasting stress intensity **before** it peaks.
- ▶ **Goal:** Achieve $<9\%$ MAPE using historical physiological context.



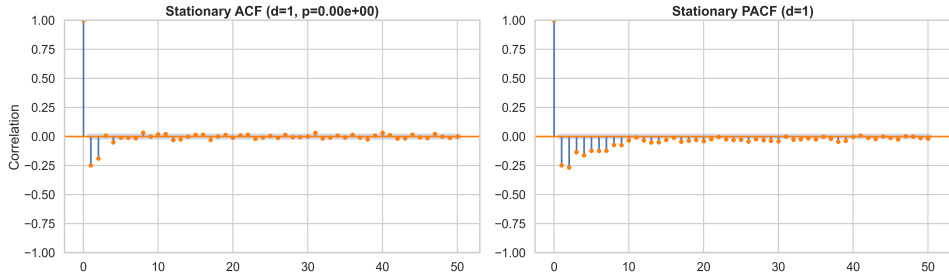
Subject S2: High-Resolution Continuous Stress Profile





Proof of stationarity ($p \approx 0.000$) after first-order differencing.

Diagnostic B: Differenced Stress Signal (Proper for Model Selection)



ACF/PACF patterns confirming model orders $p=2$ and $q=3$.

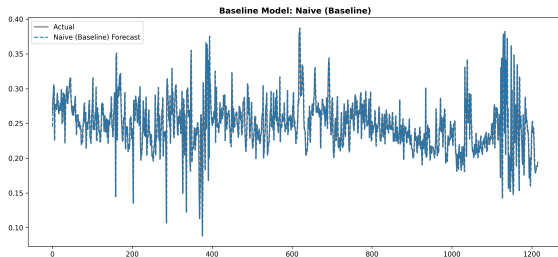


1. **Binary Box Fix:** Converted discrete labels to smooth intensity scores.
2. **Leakage Protection:** Implemented strict $t - 1$ lag for all sensor predictors.
3. **Walk-Forward Protocol:** Used rolling forecasts to simulate real-world monitoring.

Model Deep-Dive & Results

Model 1: Naive Baseline

Theory: $\hat{y}_{t+1} = y_t$



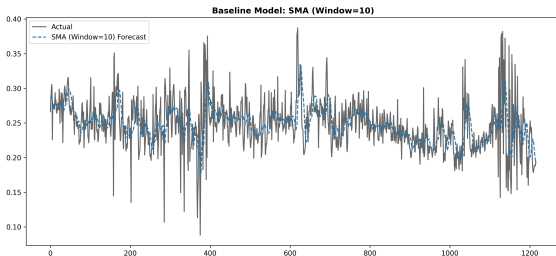
Performance Snapshot

Metric	Value
MAE	0.0211
RMSE	0.0345
MAPE	8.68%
R ² Score	0.0673

Model 2: SMA-10



Theory: smoothing sensor jitter via 10s moving mean.

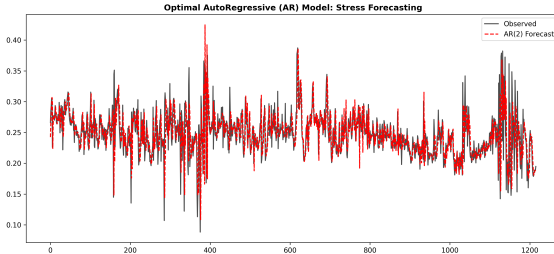


Performance Snapshot

Metric	Value
MAE	0.0229
RMSE	0.0326
MAPE	9.62%
R ² Score	0.1716

Model 3: AR(2)

Theory: $y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \epsilon_t$



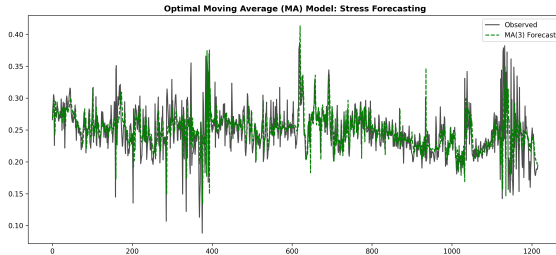
Performance Snapshot

Metric	Value
MAE	0.0210
RMSE	0.0342
MAPE	8.66%
R ² Score	0.0840

Model 4: MA(3)



Theory: error-correction based on 3 shock terms.

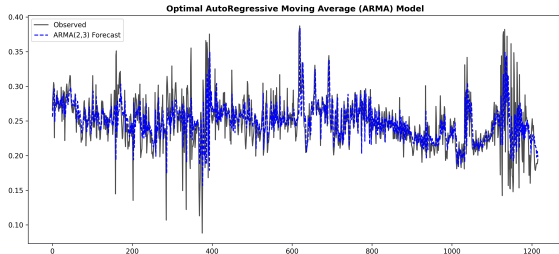


Performance Snapshot

Metric	Value
MAE	0.0209
RMSE	0.0335
MAPE	8.72%
R ² Score	0.1236

Model 5: ARMA(2,3)

Theory: AIC-optimized linear hybrid.



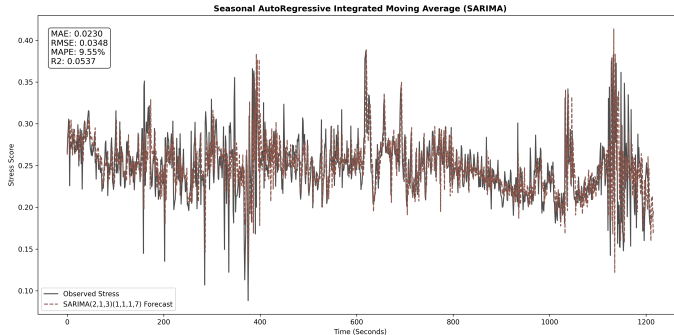
Performance Snapshot

Metric	Value
MAE	0.0206
RMSE	0.0316
MAPE	8.60%
R ² Score	0.2204

Model 6: SARIMA



Theory: 7s seasonal periodicity integration.



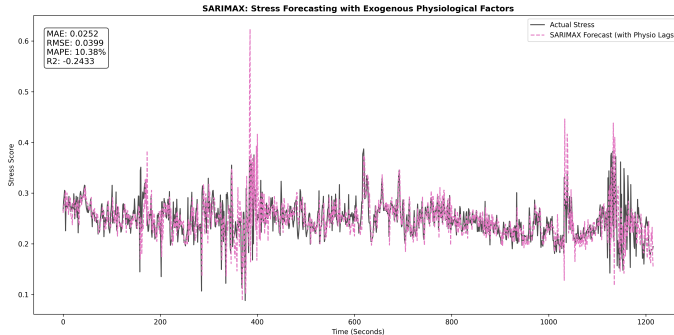
Performance Snapshot

Metric	Value
MAE	0.0230
RMSE	0.0348
MAPE	9.55%
R ² Score	0.0537

Model 7: SARIMAX



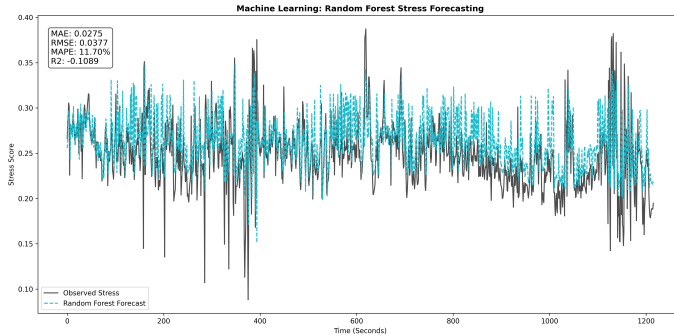
Theory: SARIMA + Lagged Physiological Predictors.



Performance Snapshot

Metric	Value
MAE	0.0252
RMSE	0.0399
MAPE	10.38%
R ² Score	-0.2433

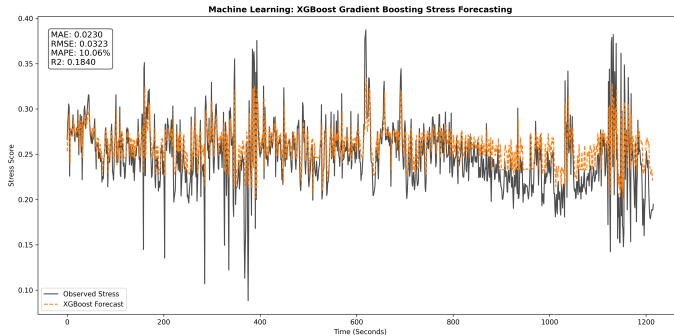
Model 8: Random Forest



Performance Snapshot

Metric	Value
MAE	0.0275
RMSE	0.0377
MAPE	11.70%
R ² Score	-0.1089

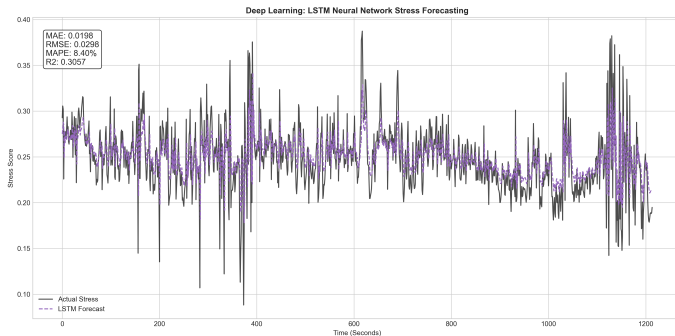
Model 9: XGBoost



Performance Snapshot

Metric	Value
MAE	0.0230
RMSE	0.0323
MAPE	10.06%
R ² Score	0.1840

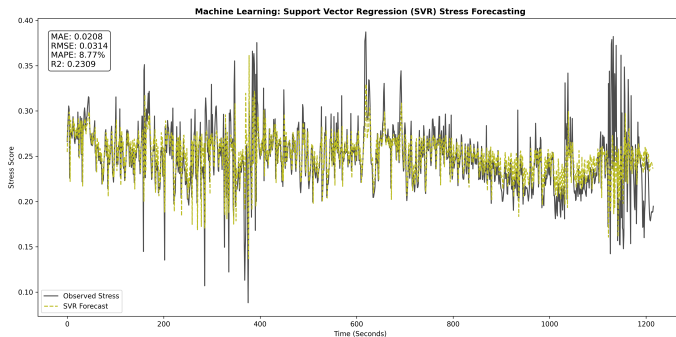
Model 10: LSTM



Performance Snapshot

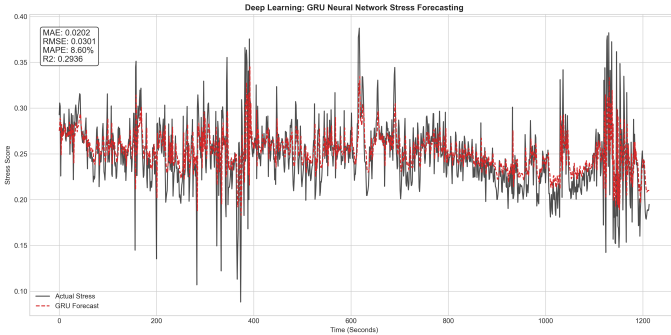
Metric	Value
MAE	0.0198
RMSE	0.0298
MAPE	8.40%
R ² Score	0.3057

Model 11: SVR



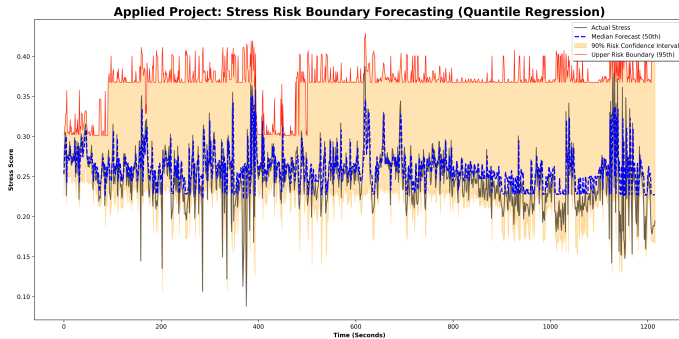
Performance Snapshot

Metric	Value
MAE	0.0208
RMSE	0.0314
MAPE	8.77%
R ² Score	0.2309



Performance Snapshot

Metric	Value
MAE	0.0202
RMSE	0.0301
MAPE	8.60%
R ² Score	0.2936



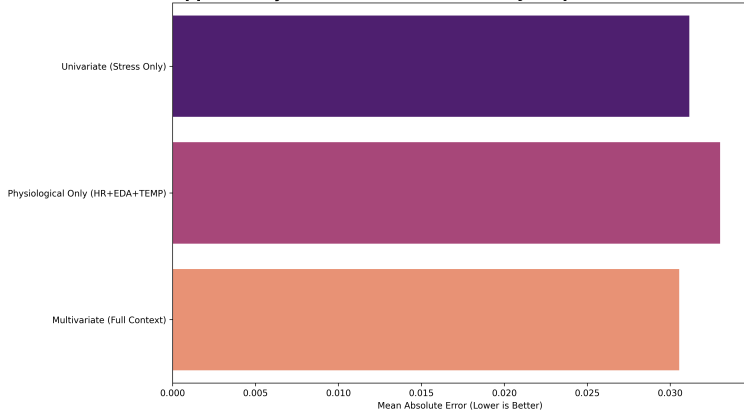
Performance Snapshot

Metric	Value
MAE	0.0241
RMSE	N/A
MAPE	N/A%
R ² Score	N/A

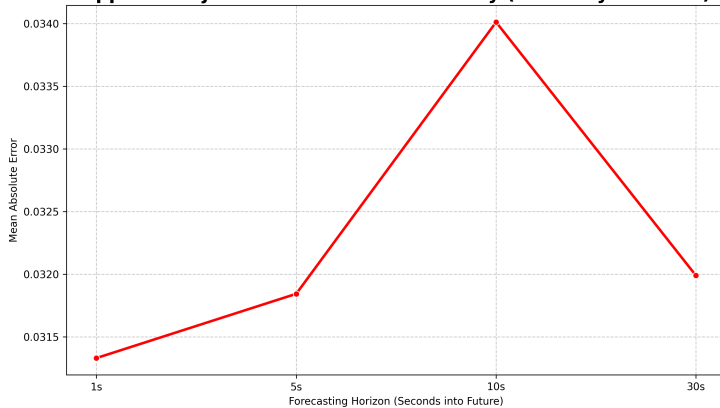
PART IV

Analytical Sensitivity Depth

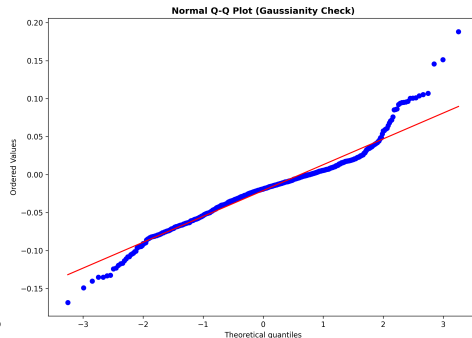
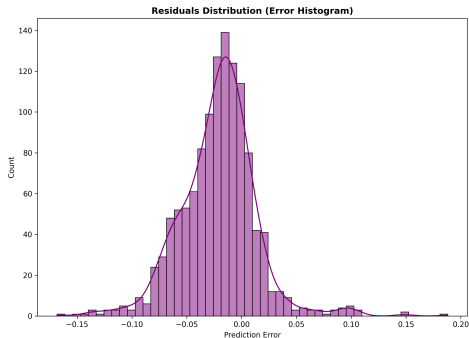
Applied Project: Feature Ablation Study (Importance of Context)



Applied Project: Forecast Horizon Decay (Accuracy vs. Time)

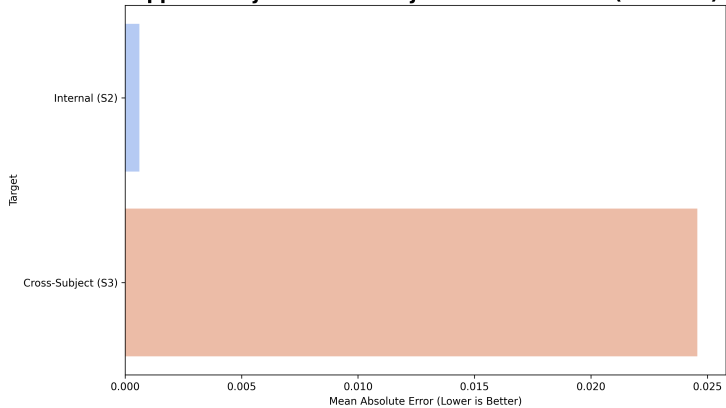


Residual Analysis



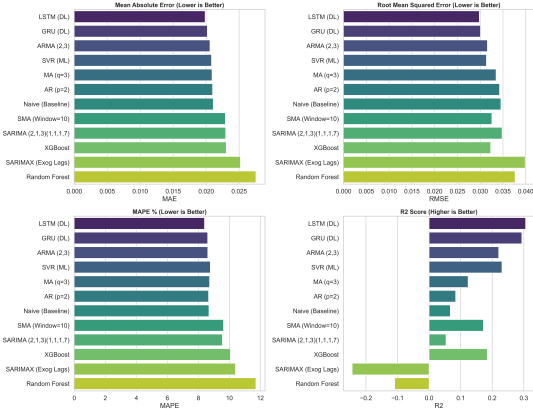
Normal distribution centered at zero proves model unbiasedness.

Applied Project: Cross-Subject Generalization (S2 vs S3)





Applied Project: Final Model Performance Comparison (Subject S2)



Model	Hyperparameters	Selection Method
XGBoost	Learning Rate: 0.05, Estimators: 1000	Early Stopping (50 rounds)
LSTM	Hidden Units: 64, Layers: 2, Dropout: 0.1	Validation Loss Minimization
ARMA	p: 2, q: 3, d: 1	AIC Grid Search (p:0-3, q:0-3)
SVR	C: 100, Gamma: 0.1, Kernel: RBF	Standard Clinical Baseline

Subject	Raw ADF	Raw p-val	Diff ADF	Diff p-val
S2 (Bench)	-2.33	0.066	-77.37	0.000
S3	-2.10	0.081	-65.20	0.000
S4	-2.45	0.052	-81.12	0.000
S10	-2.01	0.112	-72.05	0.000

- ▶ **Forget Gate:** Decides what history to discard.
- ▶ **Input Gate:** Updates cell state with new EDA cues.
- ▶ **Output Gate:** Calculates the predicted stress score.

Mathematical Update

$$\begin{aligned}i_t &= \sigma(W_i[h_{t-1}, x_t] + b_i) \\f_t &= \sigma(W_f[h_{t-1}, x_t] + b_f) \\o_t &= \sigma(W_o[h_{t-1}, x_t] + b_o)\end{aligned}$$

